

Recent Advances of Multimodal Continual Learning: A Comprehensive Survey

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Abstract—Continual learning (CL) aims to empower machine learning (ML) models to learn continually from new data, while building upon previously acquired knowledge without forgetting. As models have evolved from small to large pretrained architectures, and from supporting unimodal to multimodal (MM) data, multimodal CL (MMCL) methods have recently emerged. The primary complexity of MMCL is that it extends beyond a simple stacking of unimodal CL methods. Such straightforward approaches often suffer from MM catastrophic forgetting, yielding unsatisfactory performance. In addition, MMCL introduces new challenges that unimodal CL methods fail to adequately address, including modality imbalance, complex modality interaction, high computational costs, and degradation of the pretrained zero-shot capability of MM backbones. In this work, we present the *first* comprehensive survey on MMCL. We provide essential background knowledge and MMCL settings, as well as a structured taxonomy of MMCL methods. We categorize MMCL methods into four categories, i.e., regularization-based, architecture-based, replay-based, and prompt-based methods, explaining their methodologies and highlighting their key innovations. Additionally, to prompt further research in this field, we summarize open MMCL datasets and benchmarks, provide an in-depth discussion, and discuss several promising future directions. We have also created a GitHub repository for indexing relevant MMCL articles and open resources available at <https://github.com/LucyDYu/Awesome-Multimodal-Continual-Learning>.

Index Terms—Incremental learning, lifelong learning, multimodal continual learning (MMCL), multimodal (MM) data.

NOMENCLATURE

Notation	Description
t	Task-ID; $t \in \mathcal{T} = \{1, 2, \dots, T\}$; $T \in \mathbb{N}$ represents the total number of tasks.
$\mathcal{D}_t$	Dataset of the t th task.
$\mathcal{D}$	Entire dataset; $\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2 \cup \dots \cup \mathcal{D}_T = \bigcup_{t=1}^T \mathcal{D}_t$.

Received 20 January 2025; revised 27 July 2025; accepted 18 January 2026. Date of publication 26 March 2026; date of current version 2 September 2026. This work was supported in part by the Research Grants Council of Hong Kong Special Administrative Region, China, under Grant CUHK 2410072, RGC R1015-23 and Grant CUHK 2300246, RGC C1043-24G; and in part by NSF under Grant III-2106758 and Grant POSE-2346158. (Corresponding author: Irwin King.)

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Digital Object Identifier 10.1109/TNNLS.2026.3658485

$\mathcal{X}_t$	Input data of the t th task.
$\mathcal{X}$	$\mathcal{X} = \bigcup_{t=1}^T \mathcal{X}_t$.
$p(\mathcal{X}_t)$	Distribution of $\mathcal{X}_t$.
$\mathcal{Y}_t$	Data label of the t th task.
$\mathcal{Y}$	$\mathcal{Y} = \bigcup_{t=1}^T \mathcal{Y}_t$.
θ	Trainable parameters.
θ_t^*	Optimal parameters after training on the t th task.
$\mathcal{M}$	Episodic memory.
$\mathcal{I}$	Set of all input modalities present in $\mathcal{D}$; $\mathcal{I} = \{1, 2, \dots, I\}$; $I \in \mathbb{N}$ represents the total number of modalities; $m \in \mathcal{I}$ represents the modality-ID that labels each modality.
$\mathcal{I}_t$	Set of input modalities present in $\mathcal{D}_t$; $\mathcal{I}_t \subseteq \mathcal{I}$.

I. INTRODUCTION

IN RECENT years, machine learning (ML) has achieved significant advancements, contributing to the resolution of a wide range of practical problems. In conventional settings, most ML models operate within the so-called “single-episode” paradigm, being trained on *static* and *single* datasets, while evaluated under the independent and identically distributed (i.i.d.) assumption [1]. However, this “single-episode” paradigm may not equip the trained models with the capability to adapt to new data or perform new tasks, failing to align with the aspiration of developing intelligent agents for dynamically evolving environments. To address this issue, the ML community is motivated to develop *continual learning* (CL), also known as lifelong learning or incremental learning, which trains models incrementally on new tasks and maintains early knowledge without requiring full-data retraining [2], [3], [4], [5]. CL has become increasingly important in various fields and applications, including but not limited to the Internet of Things [6], website detection [7], and activity recognition [8].

The main challenge of CL is *catastrophic forgetting*: a phenomenon that when tasks are trained sequentially, training on the new task greatly disrupts performance on previously learned tasks [9], [10], as unconstrained fine-tuning drives parameters moving far from the old optimal state [11]. CL aims to develop learning systems that achieve a balance between *plasticity* for continuous knowledge acquisition and *stability* for retaining previously learned information [12], [13]. Such a process essentially imitates the

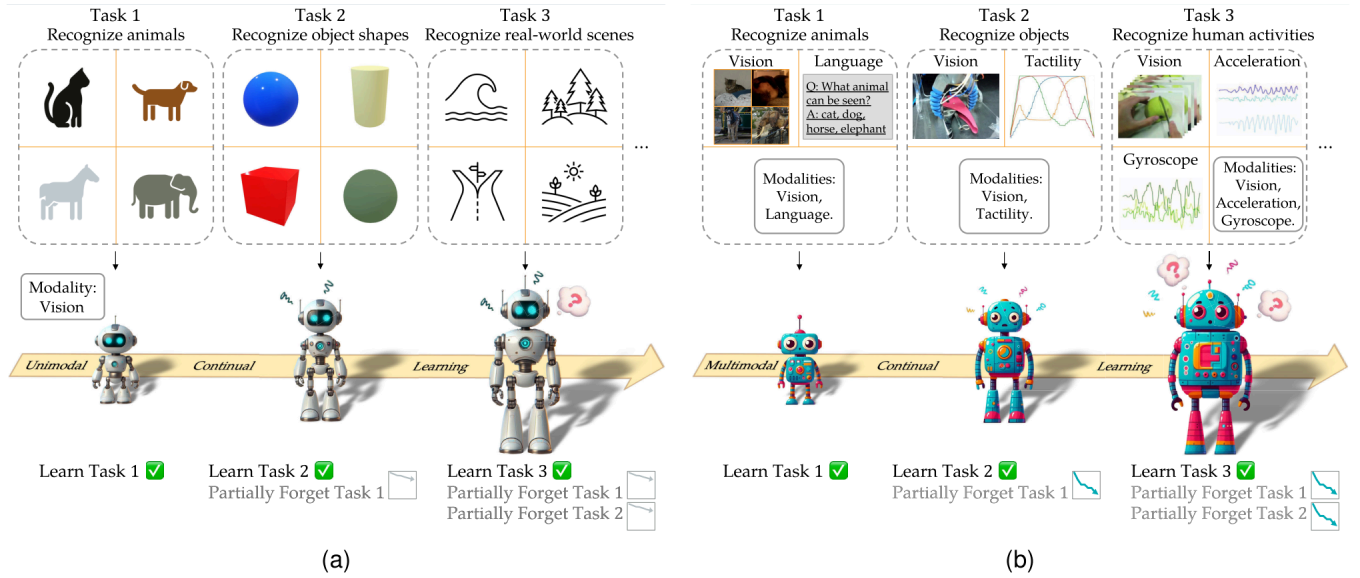


Fig. 1. Graphical illustrations of CL and MMCL. (a) Unimodal CL. The model continually learns new tasks. While learning a new task, the model tends to forget the previously learned tasks. CL aims to mitigate forgetting. (b) MMCL. In the MM setting, the model continually learns new tasks, and the dataset is MM. Forgetting in MMCL tends to be more severe due to challenges mentioned in Section 1. Example tasks in (a) are adapted based on SCD [32], VQACL [33], CLEVR [34], and GQA [35]. Example tasks in (b) are adapted based on SCD [32], VQACL [33], exFeCAM [36], and CMR-MFN [37].

cognitive flexibility observed in brains, which continually learn diverse skills throughout the human lifespan [14]. By enabling models to adapt to new tasks without forgetting, CL offers advantages in terms of resource and time efficiency compared to the traditional approach of exhaustive model retraining on full task datasets. Furthermore, due to issues of storage limitations, privacy concerns, etc., the potential inaccessibility of historical training data makes full-data training unfeasible, highlighting the efficiency and effectiveness of CL in memorizing former knowledge and acquiring up-to-date one from dynamic environments.

Despite significant progress in CL, most efforts have been devoted to a single data modality, such as vision [15], [16], [17], [18], [19], [20], language [21], [22], [23], [24], [25], or audio [26], [27]. This *unimodal focus* overlooks the multimodal (MM) nature of real-world environments, which are inherently complex and composed of diverse modalities rather than a single one. With the rapid growth of such MM data, e.g., data proliferation of images, texts, and videos on platforms like Meta and TikTok, it is imperative to develop AI systems capable of learning continually from *multimodal sources*, and thus the rise of the *multimodal CL (MMCL)* setting. These MMCL systems need to effectively integrate and process various multimodal data streams [28], [29] while managing to preserve previously acquired knowledge. More importantly, this MMCL setting better mimics the process of learning and integrating information across different modalities in human biological systems, ultimately enhancing the overall perception and cognitive capabilities when dealing with real-world complexities [30], [31]. Illustrations of unimodal CL and MMCL are provided in Fig. 1.

Challenges of MMCL: Despite the connection between conventional unimodal CL and MMCL, the challenges of

MMCL extend beyond a simple stacking of CL¹ methods on multimodal data.

Challenge 0 (Multimodal Catastrophic Forgetting): Stemming from unimodal CL, the primary and direct challenge in MMCL is multimodal catastrophic forgetting [Fig. 1(b)]. Straightforwardly applying unimodal CL methods has been demonstrated to yield suboptimal performance due to severe forgetting [39], [40], [41]. Counterintuitively, despite the availability of more data, performance with multimodal inputs may be lower than with a single modality [31], [42]. Moreover, different modalities exhibit varying forgetting rates, further complicating this issue [40].

Beyond the severe forgetting phenomenon above, the multimodal nature of MMCL also introduces the following four challenges. We detail these in the following and illustrate them in Fig. 2. These challenges not only stand alone but may also exacerbate Challenge 0.

Challenge 1 (Modality Imbalance): Modality imbalance refers to the uneven processing or representation of different modalities within a multimodal system, which manifests at both the *data* and *parameter* levels. At the data level, the data availability of different modalities may significantly vary during the CL process, with extremely imbalanced cases such as the absence of certain modalities [36]. At the parameter level, the learning of different modality-specific components may converge at varying rates, leading to a holistic imbalanced learning process across all modalities [43], [44]. This occurs because the modality with better performance may take a dominant position during optimization, whereas other

¹In this article, we use terms CL and MMCL to, respectively, refer to unimodal CL and MMCL for simplicity, if no confusion is caused.

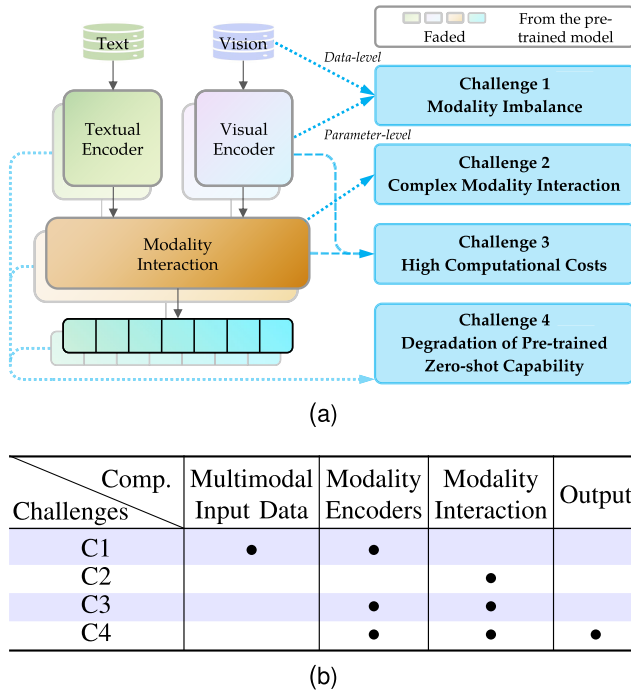


Fig. 2. MMCL challenges. We use a VLM architecture adapted from ViLT [38] as an example to illustrate. (a) Graphical illustration of model components (Comp.) and challenges. (b) Corresponding summary table of the challenge-component relationships.

modalities are under-optimized [45]. After learning multimodal knowledge, a critical consequence of imbalance is forgetting rate differences among modalities. Some modalities exhibit greater degradation than others, failing to retain old knowledge (e.g., vision degrading faster than language) [40], [46]. This imbalanced forgetting is amplified when modality-specific components encourage the model’s disproportionate focus on features from its dominant modality, while other modalities are effectively suppressed [46]. Therefore, MMCL models may suffer from performance degradation and, at times, may even perform worse than their unimodal counterparts [40], [47].

Challenge 2 (Complex Modality Interaction): Modality interaction takes place in the model components where the representations of multimodal input information explicitly interact with one another. This interaction introduces unique challenges in MMCL, primarily manifesting in two interaction processes: *modality alignment* and *modality fusion* [48]. In modality alignment, features from different modalities of a data sample tend to diverge during CL, a phenomenon known as *spatial disorder (SD) in MMCL* [49]. This divergence may cause greater performance degradation, in contrast to the more robust nature of unimodal CL, such as in image-only settings [49]. In modality fusion, a classical multimodal fusion approach used in the non-CL setting may perform worse in the MMCL setting, as different fusion techniques have varying effects on addressing the forgetting issue [42], [47]. In general, different modalities may exhibit inconsistent distributions and representations due to data heterogeneity [28], [50] and demonstrate different sensitivities to distribution

shifts [31], further complicating the alignment and fusion processes in MMCL. In addition, the uncertainties in the modality interaction stage may contribute to model overfitting on downstream tasks and knowledge forgetting [51]. Consequently, such complex modality interaction in MMCL highlights the necessity for specialized approaches to effectively incorporate multimodal data while maintaining CL capabilities.

Challenge 3 (High Computational Costs): The incorporation of multiple modalities in MMCL significantly increases computational costs at both the *model* and *task-specific* levels. At the model level, adding modalities inevitably increases the number of trainable parameters. Many MMCL methods utilize pretrained multimodal models as their foundations. However, continuously fine-tuning these large-scale models in their entirety leads to heavy computational overhead [52], [53]. At the task-specific level, similarly, MMCL methods may lead to the consistent accumulation of task-specific trainable parameters, which can potentially exceed the number of parameters in the backbone model, thereby negating the original efficiency benefits of employing CL approaches [51]. These escalating computational demands pose strict requirements on the scalability of MMCL methods for practical deployment, especially given resource constraints.

Challenge 4 (Degradation of Pretrained Zero-Shot Capability): With advances in pretrained models, MMCL methods can be armed with these powerful foundations. Consequently, these pretrained multimodal models exhibit *zero-shot* capability on unseen tasks [54], [55], which distinguishes MMCL methods from traditional unimodal CL methods that usually train from scratch. However, during continuous fine-tuning of MMCL, some initial capabilities derived from pretraining foundations, such as performing zero-shot tasks, may diminish. Such degradation risk may lead to severe performance decay on future tasks [55], known as *negative forward transfer in MMCL* [54]. This phenomenon highlights that MMCL approaches must maintain the delicate balance between retaining pretrained capabilities and adapting to new tasks.

Contributions: To address these challenges, researchers are increasingly focusing on MMCL methodologies. As detailed in Section III, our taxonomy categorizes the MMCL methods into four main approaches, i.e., regularization-based, architecture-based, replay-based, and prompt-based methods. Given the increasing importance and research interest in MMCL, we present the *first* comprehensive MMCL survey. Beyond discussing the challenges faced in MMCL, this article systematically details the basic formulations and settings (Section II), reviews existing methodologies (Section III), summarizes relevant datasets and benchmarks (Section IV), provides a thorough discussion of the field (Section V), and outlines promising future directions (Section VI). Our goal is not only to consolidate current MMCL advancements but also to inspire innovative research, thereby fostering the development of more effective and efficient MMCL approaches. In summary, this article makes the following key contributions.

- 1) We present the first comprehensive survey on MMCL. We start by detailing essential MMCL background

knowledge, including the basic formulation, distinct MMCL scenarios, and widely used evaluation metrics.

- 2) In our structured taxonomy of MMCL methods, we categorize existing MMCL works into four categories with thorough subcategory explanations. For each category, we provide representative architecture illustrations and offer a detailed methodology review, highlighting key features and innovations accordingly.
- 3) We summarize the current datasets and benchmarks to facilitate research and experiments. For this rapidly evolving field of MMCL, we provide an in-depth discussion of research trends, foundation models in MMCL, and the evolved stability–plasticity tradeoff. We discuss promising future research directions, providing insights into potential areas for further investigation and development.

Connections With Other CL Surveys: Several surveys are available mainly for general CL methodologies [5], [56], [57]. There are also CL surveys focusing on the specific unimodal modality, such as computer vision [13], [14] and natural language processing [58], [59], [60]. Additionally, with the advancement of pretrained models and foundation language models, two works specifically review these developments [61], [62]. There are CL surveys that specifically focus on other aspects like federated learning (FL) settings [63] and knowledge distillation (KD) [64], but they do not discuss MMCL methods. This work aims to present a comprehensive MMCL survey, addressing the lack of a dedicated survey in this area and filling this gap.

II. PRELIMINARIES

In this section, we introduce the setup for MMCL, including notations, basic formulation, distinct learning scenarios, and widely used evaluation metrics.

A. Notations

We use bold lowercase, bold uppercase, and calligraphy letters for vectors, matrices, and sets, respectively. We list the key notations in the Nomenclature. The notations of $\mathcal{X}_t$, $p(\mathcal{X}_t)$, $\mathcal{Y}_t$, and $\mathcal{T}$ are adopted from [5].

B. Basic Formulation

In this section, we introduce the basic formulation of CL and MMCL. Definitions 1 and 2 define task sequence and CL, respectively. Then, we present the first formal definitions related to MMCL in Definitions 3–8.

Definition 1 (Task Sequence): Let $\mathcal{X}_t$ and $\mathcal{Y}_t$ denote the input data and the data label of the t th task, respectively. The dataset of the t th task, denoted as $\mathcal{D}_t$, is defined as

$$\mathcal{D}_t = \{(\mathbf{x}_{t,i}, y_{t,i}) : i \in \mathbb{N}, 1 \leq i \leq N_t\} \quad (1)$$

where $\mathbf{x}_{t,i} \in \mathcal{X}_t$ and $y_{t,i} \in \mathcal{Y}_t$ are the i th data, and N_t is the number of samples of the t th task. A *task sequence* TS of size T (where $T > 1$ is required) is a sequence of tasks with their datasets in a certain order, defined as

$$TS = [\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_T]. \quad (2)$$

We define the set of task-IDs as $\mathcal{T} = \{1, 2, \dots, T\}$. $\forall t \in \mathcal{T}, TS[t] = \mathcal{D}_t$.

Definition 2 (CL): Given a task sequence TS of size T , we consider the t th task ($1 < t \leq T$) as a new task so far. *CL* is the setting that, for each such task, the model is trained only on data $\mathcal{D}_t$ (or with very limited access to previous datasets $\{\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_{t-1}\}$ in a more relaxed setting). The objective is to learn the new task while maintaining performance on old tasks to overcome catastrophic forgetting. Specifically, given an unseen test sample $\mathbf{x} \in \mathcal{X}$ from any trained tasks, the trained model $f : \mathcal{X} \rightarrow \mathcal{Y}$ should perform well in inferring the label $y = f(\mathbf{x}) \in \mathcal{Y}$ [65].

Remark 1: The performance of the model is evaluated using metrics described in Section II-D. The difficulty of CL stems from the fact that datasets have dynamic distributions, i.e., $\forall i, j \in \mathcal{T}, i \neq j \Rightarrow p(\mathcal{X}_i) \neq p(\mathcal{X}_j)$ [5].

Definition 3 (Modality-IDs and Set): Let $\mathcal{D}$ be the union of datasets of all tasks, defined as $\mathcal{D} = \bigcup_{t=1}^T \mathcal{D}_t$. Let I be the total number of input modalities (e.g., vision, language, audio, etc.) present in $\mathcal{D}$. We define the *modality set* $\mathcal{I} = \{1, 2, \dots, I\}$. Let $m \in \mathcal{I}$ represent the *modality-ID*, labeling each modality as a mathematical abstraction. $\forall t \in \mathcal{T}$, let $\mathcal{I}_t$ be the *modality set* of $\mathcal{D}_t$, $\mathcal{I}_t \subseteq \mathcal{I}$.

Definition 4 (Unimodal and Multimodal): Given a task sequence TS of size T , we say the following.

- 1) TS is *unimodal* if $|\mathcal{I}| = 1$; i.e., $\mathcal{D}$ contains one modality.
- 2) TS is *multimodal* if $|\mathcal{I}| > 1$; i.e., $\mathcal{D}$ contains more than one modality.

Definition 5 (Modality-Static, Modality-Increasing, Modality-Decreasing, and Modality-Switching): Given a task sequence TS of size T , we say the following.

- 1) TS is *modality-static* if $\forall i, j \in \mathcal{T}, \mathcal{I}_i = \mathcal{I}_j$; i.e., all datasets have the same modality (or modalities).
- 2) TS is *modality-increasing* if $\forall 1 < i \leq T, \mathcal{I}_{i-1} \subsetneq \mathcal{I}_i$; i.e., for each new task, it has more modalities compared to the previous task.
- 3) TS is *modality-decreasing* if $\forall 1 < i \leq T, \mathcal{I}_{i-1} \supsetneq \mathcal{I}_i$; i.e., for each new task, it has fewer modalities compared to the previous task.
- 4) TS is *modality-switching* if $\forall 1 < i \leq T, (\mathcal{I}_{i-1} \not\subseteq \mathcal{I}_i) \wedge (\mathcal{I}_{i-1} \not\supseteq \mathcal{I}_i)$; i.e., two consecutive tasks have different modalities and do not have a subset relationship. For each new task, it switches modalities to involve different ones compared to the previous task.

Definition 6 (Subsequence): Given a task sequence TS of size T , a task sequence TS' of size T' is a *subsequence* of TS if $\exists i \in \mathbb{N} \forall t \in \{1, 2, \dots, T'\}, TS'[t] = TS[t + i]$ (or equivalently, $\exists i \in \mathcal{T}, TS' = TS[i : i + T'] = [\mathcal{D}_i, \mathcal{D}_{i+1}, \dots, \mathcal{D}_{i+T'-1}]$).

Definition 7 (Modality-Dynamic): A multimodal task sequence TS is *modality-dynamic* if it has a subsequence that is modality-increasing, modality-decreasing, or modality-switching.

Definition 8 (MMCL): Given a task sequence TS , *MMCL* is the setting where TS is multimodal, and the model is trained under the CL setting.

Remark 2: In addition to the challenge of catastrophic forgetting present in CL, MMCL introduces four challenges

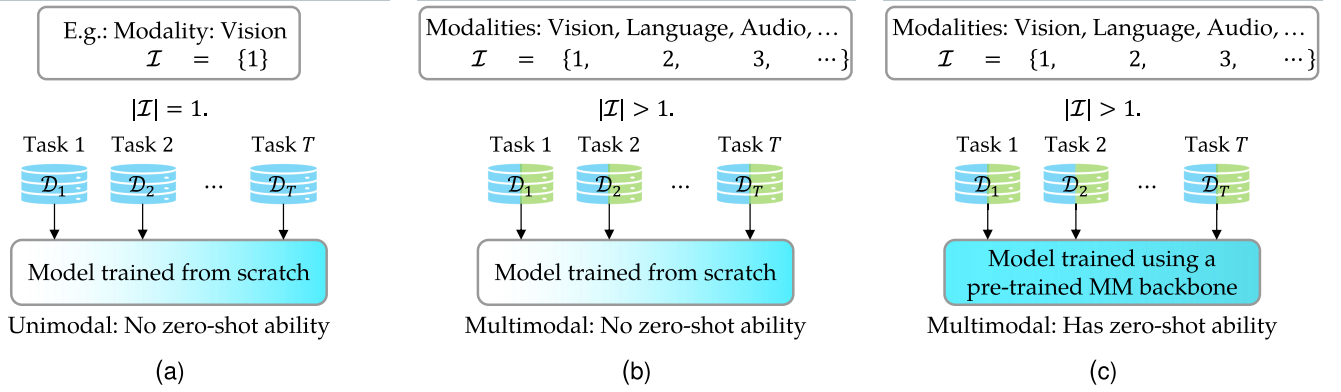


Fig. 3. Illustrations of CL and MMCL. Notations are defined in the Nomenclature. (a) Unimodal CL. The model is trained from scratch. (b) MMCL. The model is trained from scratch. (c) MMCL. The model is trained using a pretrained MM backbone.

as described in Section I. Moreover, when the task sequence is modality-dynamic, MMCL demonstrates increased flexibility but introduces greater complexity (see [36], [39], [66]).

Fig. 3 provides graphical illustrations of CL and MMCL. Fig. 3(a) illustrates the case when in conventional CL, datasets have the same single modality, and the model is trained from scratch. The model lacks the zero-shot ability. With multimodal datasets, MMCL methods may be trained either from scratch [Fig. 3(b)] or using a pretrained MM backbone [Fig. 3(c)]. A model with a pretrained MM backbone possesses zero-shot ability, i.e., to give zero-shot predictions on tasks. For example, the pretrained CLIP model [67] achieves zero-shot image classification accuracy of 88.5% and 89.0% on datasets of Food [68] and OxfordPet [69], respectively [55]. MMCL methods that use pretrained MM backbones should address Challenge 4 to preserve zero-shot capabilities throughout the learning process.

C. MMCL Scenarios

In MMCL, the learning process varies in terms of modalities, data distribution, and task identity availability, resulting in five different MMCL scenarios. We first introduce three scenarios that originate in conventional CL but can be inclusive in MMCL.

Scenario 1 [Class-Incremental Learning (CIL)]: For $i \neq j$, $\mathcal{D}_i$ and $\mathcal{D}_j$ have different input distributions and data label spaces, i.e., $p(\mathcal{X}_i) \neq p(\mathcal{X}_j) \wedge \mathcal{Y}_i \neq \mathcal{Y}_j$ [5]. Task identities are not available in testing. The model should be able to perform classification for all seen classes. The model may need to infer the task-ID at test time to determine the possible classes of a test sample [4]. Note that in the conventional CIL setting, the data label spaces of tasks are disjoint, i.e., $\forall i \neq j, \mathcal{Y}_i \cap \mathcal{Y}_j = \emptyset$ [5]; however, in a more generalized CIL setting, the data label spaces may overlap, i.e., $\exists i \neq j, \mathcal{Y}_i \cap \mathcal{Y}_j \neq \emptyset$ [31].

Scenario 2 [Domain-Incremental Learning (DIL)]: For $i \neq j$, $\mathcal{D}_i$ and $\mathcal{D}_j$ have different input distributions but the same label space, i.e., $p(\mathcal{X}_i) \neq p(\mathcal{X}_j) \wedge \mathcal{Y}_i = \mathcal{Y}_j$ [5]. Task identities are not required. Identifying the task is unnecessary for the model because of the same label space of all tasks [4].

Scenario 3 [Task-Incremental Learning (TIL)]: For $i \neq j$, $\mathcal{D}_i$ and $\mathcal{D}_j$ have different input distributions and label spaces, i.e., $p(\mathcal{X}_i) \neq p(\mathcal{X}_j) \wedge \mathcal{Y}_i \neq \mathcal{Y}_j$ [5]. Task identities are available in testing. The model needs to learn the tasks, and with the task-ID received at test time, it knows which task needs to be performed [4].

For MMCL, we introduce two new scenarios.

Scenario 4 [Cross-DIL (XDIL)]: For $i \neq j$, $\mathcal{D}_i$ and $\mathcal{D}_j$ have different input distributions and label spaces, i.e., $p(\mathcal{X}_i) \neq p(\mathcal{X}_j) \wedge \mathcal{Y}_i \neq \mathcal{Y}_j$. Task identities are not required. This is a new scenario in MMCL, for tasks such as generative visual question answering (VQA) [33], [73], and cross-domain classification [76], [77] based on pretrained vision-language models (VLMs). The fundamental distinction between traditional CIL and XDIL lies in the dataset label spaces and model outputs. In CIL, model predictions correspond to labels in the seen datasets. However, in XDIL, the model outputs predictions from a larger set $\mathcal{Y}'$, e.g., the vocabulary set for VQA and the union set of all seen and unseen classes for classification. Labels of $\mathcal{D}_i$ are a subset of this predefined large set, $\mathcal{Y}_i \subset \mathcal{Y}'$. We view $\mathcal{Y}'_i := \mathcal{Y}'$ as the actual label space. As such, the model's effective output label space remains consistent for all datasets, i.e., $\mathcal{Y}'_i = \mathcal{Y}'_j$, although input datasets have different label spaces. From the model's perspective, this consistency makes this scenario a domain incremental variant. However, with a larger label space, it is more challenging and realistic than conventional CIL and DIL [76]. We name it XDIL to emphasize the unique relationship between the input domains and the label space.

Scenario 5 [Modality-Dynamic TIL (MDTIL)]: In unimodal CL, the task sequence is naturally modality-static (Definition 5) since there is only one modality. While in MMCL, on the one hand, if the task sequence is modality-static, it falls into one of the four scenarios described above. On the other hand, the datasets may have different modalities, i.e., $\exists i, j \in \mathcal{T}, \mathcal{I}_i \neq \mathcal{I}_j$, and the task sequence is modality-dynamic (Definition 7, see [36], [39], [66]). For $i \neq j$, $\mathcal{D}_i$ and $\mathcal{D}_j$ have different input distributions and label spaces, i.e., $p(\mathcal{X}_i) \neq p(\mathcal{X}_j) \wedge \mathcal{Y}_i \neq \mathcal{Y}_j$. Task identities are available in testing

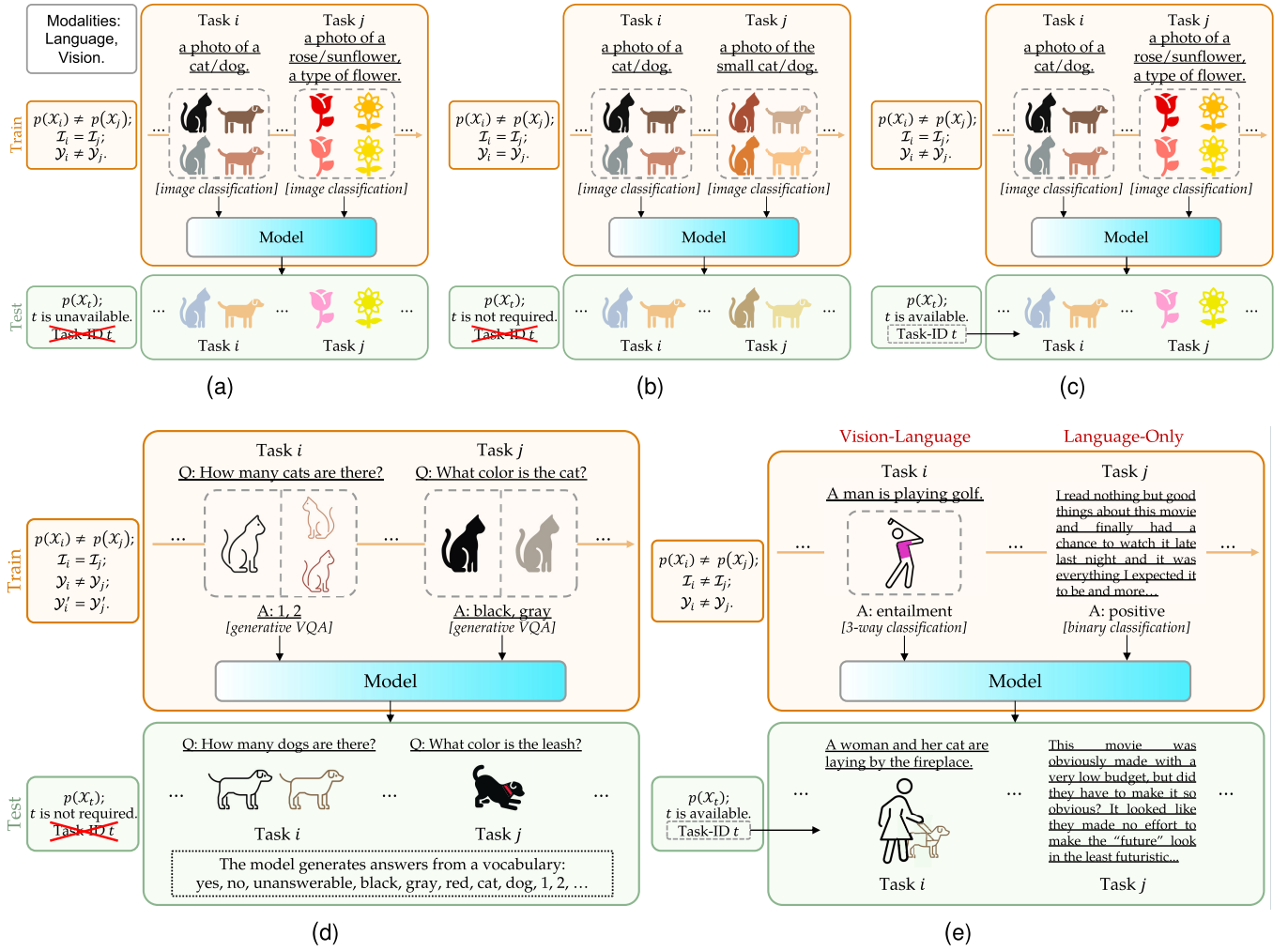


Fig. 4. Illustrations of MMCL scenarios (defined in Section II-C). Notations are defined in the Nomenclature. (a) CIL. (b) DIL. (c) TIL. (d) XDIL. (e) MDTIL. (a)–(c) are partially adapted and redrawn based on [13] and [62], with examples adapted based on MTIL [55], CIFAR10 [70], and Flowers [71]. Examples in (d) are adapted based on VQAv2 [72], VQACL [33], and SGP [73]. Examples in (e) are adapted based on CLiMB [39], SNLI-VE [74], and IMDb [75].

and are required to handle tasks with different input modalities [36], [39], [66]. We name this scenario as MDTIL.

We provide illustrations of all these five MMCL scenarios in Fig. 4. We use vision and language tasks as examples for illustration purposes, but MMCL scenarios can include tasks of various other modalities.

D. Evaluation Metrics

To evaluate the model performance in MMCL, various metrics are proposed. In the single-task case, the performance evaluation metrics may vary depending on different task types. For instance, these metrics may include accuracy for classification [52], BLEU-4 for text generation [82], and recall for retrieval tasks [80]. Based on these single-task evaluation metrics, we introduce common metrics for multiple tasks. Let $a_{t,i} \in [0, 1]$ be the model performance on the test set of the i th task, after the model is trained progressively from task 1 to task t [5]. Similarly, let $a_{t,i}^{(m)}$ be the performance using only modality m , if applicable.

- 1) *Average Performance (A)*: The average performance at the t th task is defined as

$$A_t = \frac{1}{t} \sum_{i=1}^t a_{t,i}. \quad (3)$$

- 2) *Forgetting Measures (F)* [97]: Forgetting is quantified as the difference between the “maximum” knowledge and the current knowledge of a task during the CL process. Let $f_i^t \in [-1, 1]$ be the forgetting measure of the i th task ($i < t$), after the model is progressively trained from task 1 to task t

$$f_i^t = \max_{s \in \{1, \dots, t-1\}} a_{s,i} - a_{t,i}, \forall i < t. \quad (4)$$

The average forgetting at the t th task is defined as

$$F_t = \frac{1}{t-1} \sum_{i=1}^{t-1} f_i^t. \quad (5)$$

- 3) *Backward Transfer (BWT)* [5], [98]: The difference $a_{t,i} - a_{i,i}$ measures the influence of a task t on a previous task

i ($i < t$). For the t th task, BWT measures its average influence on the performance of all previous tasks

$$\text{BWT}_t = \frac{1}{t-1} \sum_{i=1}^{t-1} a_{t,i} - a_{i,i}. \quad (6)$$

- 4) *Forward Transfer (FWT)* [98]: Let $\bar{b}_i$ be the performance of the i th task with random initialization. The difference $a_{i,t} - \bar{b}_i$ measures the influence of a task i on a future task t ($i < t$). For the t th task, FWT is defined as

$$\text{FWT}_t = \frac{1}{t-1} \sum_{i=2}^t a_{i-1,i} - \bar{b}_i. \quad (7)$$

While traditional CL metrics provide general assessments, MMCL methods propose new metrics that offer targeted analysis relating to MMCL challenges.

- 5) *Zero-Shot Transfer* [55]: A recent MMCL method, zero-shot continual learning (ZSCL), proposes the “Transfer” metric to measure the level of preserved zero-shot ability (Challenge 4) on a task t ($t > 1$), after training on tasks before it. For the t th task

$$\text{Transfer}_t = \frac{1}{t-1} \sum_{i=1}^{t-1} a_{i,t}. \quad (8)$$

- 6) *Average Performance over Modalities (AM)* [91]: For multimodal methods that support training and inference for single modality input [31], [91], the AM metric provides a way to assess performance across individual modalities. Proposed by exFeCAM, it is a modality-oriented metric, unlike conventional CL metrics. Let $A_t^{(m)}$ be the average performance on modality m at the t th task. For the t th task, AM is the average over all modalities $\mathcal{I}$, defined as

$$\text{AM}_t = \frac{1}{I} \sum_{m \in \mathcal{I}} A_t^{(m)}. \quad (9)$$

- 7) *Average Difference (AD)* [46]: To explicitly quantify modality imbalance (Challenge 1), MM-Prompt proposes the AD metric. It measures the performance gap between the best and worst modalities. For the t th task

$$\text{AD}_t = \frac{1}{t} \sum_{i=1}^t \max_{m \in \mathcal{I}} a_{t,i}^{(m)} - \min_{m \in \mathcal{I}} a_{t,i}^{(m)}. \quad (10)$$

Lower AD values indicate better balance among modalities, making the effectiveness of different methods addressing Challenge 1 comparable.

- 8) *Average Merge Effectiveness (AME)* [46]: MM-Prompt proposes the AME metric to measure the contribution of using multimodal data over the best unimodal data. For the t th task

$$\text{AME}_t = \frac{1}{t} \sum_{i=1}^t a_{t,i} - \max_{m \in \mathcal{I}} a_{t,i}^{(m)}. \quad (11)$$

Higher AME values indicate a greater performance gain from multimodal data, reflecting the method’s capability to resolve Challenge 2.

III. METHODOLOGY

In this section, we present a taxonomy of MMCL methods. Fig. 5 categorizes MMCL methods into four types, which we elaborate in Sections III-A–III-D. We summarize detailed properties of MMCL methods in Table I and the representative architectures of MMCL methods in Fig. 6. Note that Table I and Fig. 6 focus on methods of vision and language modalities, and methods of other modalities are summarized in Table II.

A. Regularization-Based Approach

Since the free movement of parameters in training causes catastrophic forgetting [11], regularization-based methods are motivated to add constraints on the parameters. Let $\mathcal{L}_{\text{single},t}$ be the loss in the single task setting when the model is learning the t th task. The continual loss is then defined as $\mathcal{L}_t = \mathcal{L}_{\text{single},t} + \lambda \mathcal{L}_{R,t}$, where $\mathcal{L}_{R,t}$ is the regularization term and the hyperparameter λ balances plasticity on the new task with stability for old tasks. Based on the purpose of regularization for preserving multimodal information, regularization-based methods are divided into two subdirections: *modality consistency regularization* and *modality relation regularization*. We summarize the representative architectures of the above two subcategories in Fig. 6(a).

1) *Modality Consistency Regularization*: Modality consistency regularization aims to maintain the consistency of learned modality-specific representations or fused information of previous tasks, while learning new tasks. This category primarily aims to directly address *Challenge 0*. One approach to preserve consistency is to regularize at the *parameter-level*, by directly assigning importance to parameters and penalizing them differently when they deviate from the previously found solution. $\mathcal{L}_{R,t}$ can be formulated as follows [16]:

$$\mathcal{L}_{R,t} = \sum_i b_i (\theta_i - \theta_{t-1,i}^*)^2 \quad (12)$$

where θ_i and $\theta_{t-1,i}^*$ denote the i th element in θ and θ_{t-1}^* , respectively, and b_i indicates the corresponding importance.

Instead of assigning weights to individual parameters, another approach keeps modality information consistent by minimizing changes in the model’s output on past knowledge, i.e., regularizing at the *output-level*. Specifically, this output may be *modality-specific features* from their respective encoders, or fused representations and model logits, which contain *multimodal fused information*. Unlike parameter-level regularization, output-level regularization offers greater flexibility in the design of the constraint. These methods typically employ *logits-based* and *feature-based* KD [99], which matches the output between a teacher model (the previous-task trained model) and a student model (the current model) [5]. Therefore, KD encourages the student model to learn logits and feature distribution similar to those of the teacher model [100], [101]. $\mathcal{L}_{R,t}$ incorporates KD and can be formulated as follows [15], [79]:

$$\begin{aligned} \mathcal{L}_{R,t} &= \mathcal{L}_{\text{KD}}(\mathbf{y}_{t-1}, \mathbf{y}_t) \\ &= \begin{cases} -\sum_i y_{t-1,i} \log y_{t,i}, & \text{cross-entropy loss} \\ \|\mathbf{y}_{t-1} - \mathbf{y}_t\|_2^2, & \text{L2 loss} \end{cases} \end{aligned} \quad (13)$$

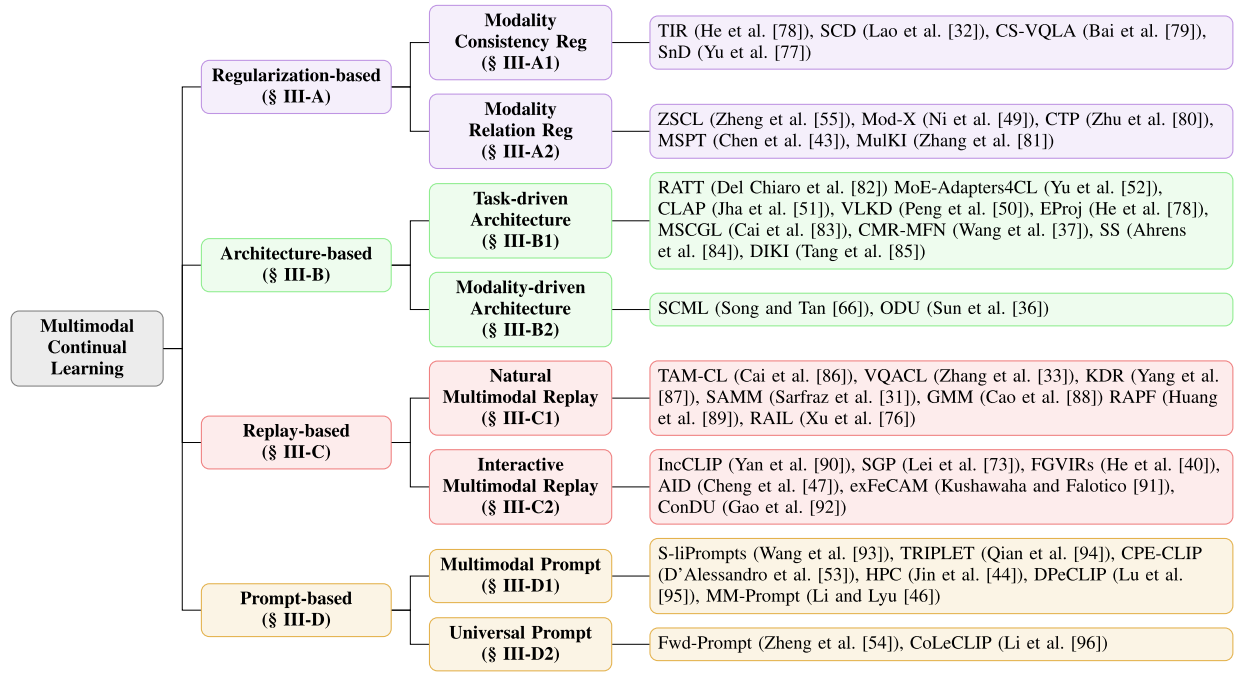


Fig. 5. Taxonomy of MMCL. We divide MMCL methods into four categories: regularization-based (Section III-A), architecture-based (Section III-B), replay-based (Section III-C), and prompt-based (Section III-D).

where $\mathbf{y}_{t-1}$ and $\mathbf{y}_t$ are the outputs of one data sample from the model before and after training on the t th task. $y_{t-1,i}$ and $y_{t,i}$ denote the i th element in $\mathbf{y}_{t-1}$ and $\mathbf{y}_t$, respectively.

a) *MMCL Methods*: In the MMCL setting, methods propose modality consistency regularization algorithms built upon multimodal base models. For example, SnD [55] utilizes CLIP [67], while TIR [78] leverages multimodal large language models (MLLMs) such as BLIP2 [102] and InstructBLIP [103]. These MMCL methods utilize regularization techniques at either the parameter-level or the output-level. Therefore, we present methods based on the above two levels for the following explanations.

At the *parameter-level*, TIR [78] measures parameter importance using existing methods such as EWC [16]. It calculates task similarity scores between new and old tasks to obtain adaptive weights for regularization, facilitating long-term CL.

At the *output-level*, CS-VQLA [79] and SCD [32] employ both *logits-based* and *feature-based* KD techniques. CS-VQLA proposes rigidity-plasticity-aware distillation (logits level) to deal with nonoverlapping and overlapping classes separately in the CIL scenario. Moreover, it introduces self-calibrated heterogeneous distillation (feature level) to minimize the distance between the self-calibrated feature map and the old feature map. SCD [32] proposes to transfer domain knowledge through self-critical distillation at both the logits and feature levels. It defines instance-relevant and domain-relevant knowledge based on teacher model prediction and proposes a self-critical temperature to adjust the knowledge transfer. SnD [77] selectively employs feature-based KD with the most recent fine-tuned and the original pretrained models as dual teachers, preserving both knowledge from past tasks and zero-shot capabilities (Challenge 4).

2) *Modality Relation Regularization*: With paired multimodal data samples, such as image-text pairs, modality relation regularization aims to preserve the learned pairwise relations between modalities from previous tasks, thereby mitigating the forgetting of cross-modal alignments. This category mainly addresses *Challenge 2*. Methods in this category typically utilize *relation-based* KD [101], which guides the current model to mimic the intermodality relations captured by the previous-task trained model. Let $\mathcal{F}^{(m_1)}$ and $\mathcal{F}^{(m_2)}$ be the feature sets of two modalities. $\mathcal{L}_{R,t}$ incorporates relation-based KD and can be formulated as follows [101]:

$$\mathcal{L}_{R,t} = \mathcal{L}_R \left(\Psi \left(\mathcal{F}_{t-1}^{(m_1)}, \mathcal{F}_{t-1}^{(m_2)} \right), \Psi \left(\mathcal{F}_t^{(m_1)}, \mathcal{F}_t^{(m_2)} \right) \right) \quad (14)$$

where $\Psi(\cdot, \cdot)$ is the relational function. Its output represents the relational knowledge, which can be a matrix (for many-to-many relations) or a vector (for one-to-many relations). $\mathcal{L}_R(\cdot)$ is the distillation loss function, such as the Frobenius norm [81] and cross-entropy loss [55].

a) *MMCL Methods*: ZSCL [55] calculates the feature similarity between each image (or text) and various texts (or images) in a reference dataset using the fine-tuned and the frozen pretrained CLIP models. It, then, employs KD to match the similarity distributions, thus preserving zero-shot transfer ability (Challenge 4). Similarly, CTP [80] and MulKI [81] use distillation to maintain the image-text similarity distribution for cross-modal topology preservation. MulKI [81] additionally distills the relation matrices of visual features to maintain intramodal relationships. Ni et al. [49] demonstrate that continual training of CLIP will cause the SD issue in vision-language representation and may lead to downgraded performance. Thus, their proposed method, Mod-X, preserves the spatial distribution of representations between modalities using distillation on the contrastive matrices of the

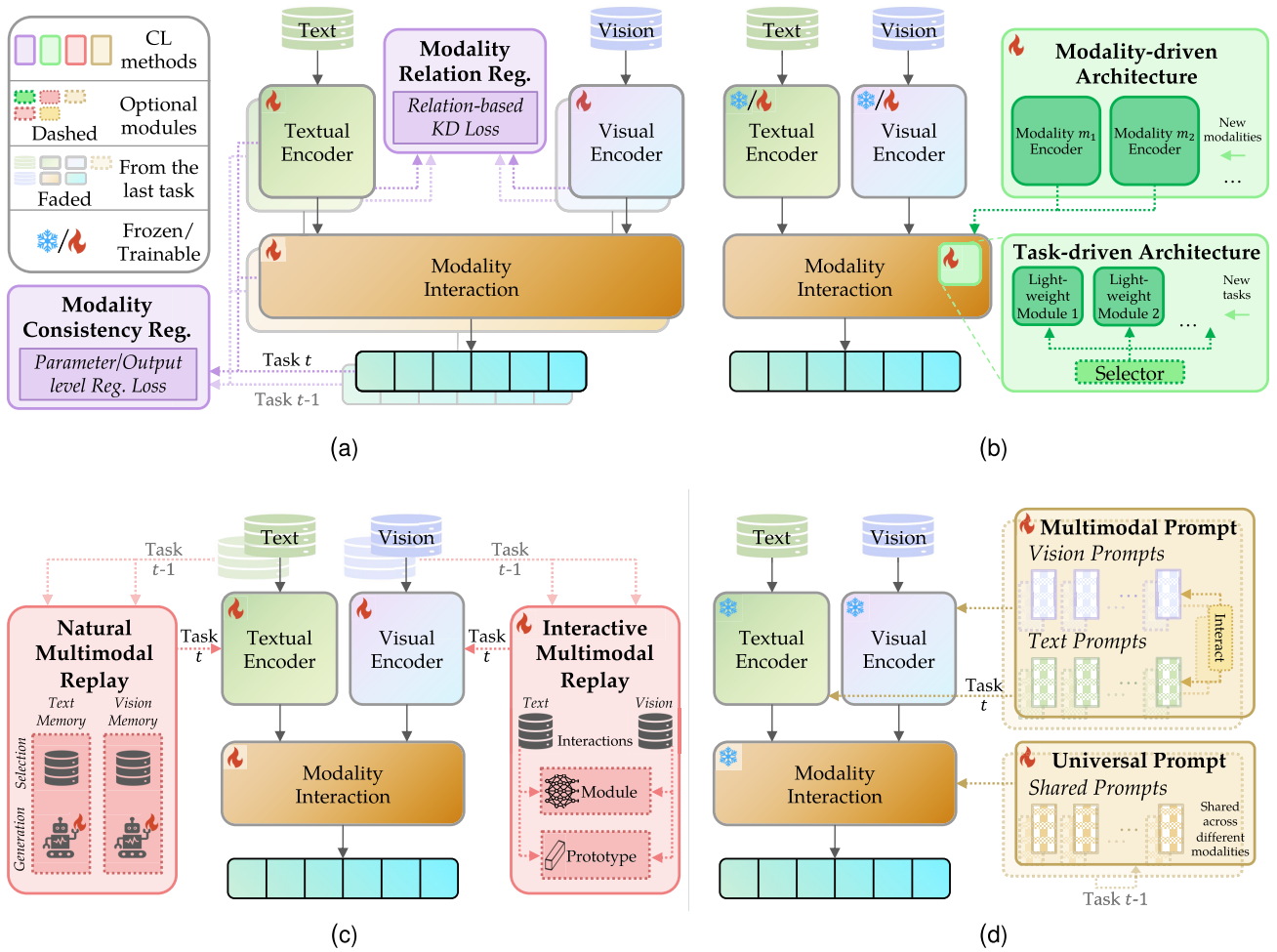


Fig. 6. Representative architectures for different categories of MMCL methods for vision and language. The base model architecture is adapted and redrawn based on ViLT [38]. The prompt-based method architecture is partially adapted and redrawn based on TRIPLET [94]. (a) Regularization-based. (b) Architecture-based. (c) Replay-based. (d) Prompt-based.

last and current CLIP models. MSPT [43] captures modality interactions via shared learnable keys in image and text self-attention modules and applies distillation to the attention maps for knowledge maintenance. To address modality imbalance (Challenge 1), it proposes a gradient modulation strategy to balance the learning of modalities, inspired by OGM-GE [45].

b) Summary with Pros and Cons: Fig. 6(a) provides a summary of typical architectures of modality consistency and relation regularization-based methods. They effectively preserve knowledge of past models with clear benefits. A primary advantage is that methods in this category are model-agnostic and easily applicable to various multimodal backbones. They typically focus on designing sophisticated loss functions rather than altering the model architecture. Furthermore, modality relation regularization methods that employ relation-based KD offer a natural fit to preserve the crucial cross-modal relationships inherent in paired data, thereby facilitating the development of various MMCL methods in this subcategory and demonstrating great potential for future research.

Despite their advantages, these methods present several limitations. First, methods usually require full-parameter fine-tuning and demand heavy computation overhead [85]

(Challenge 3). Second, methods such as ZSCL rely on external reference datasets, which pose practical challenges in real-world scenarios [85] and make performance sensitive to the choice of this dataset [81]. Lastly, we observe that the modality consistency regularization methods mainly directly address Challenge 0 and rarely tackle other MMCL challenges. This narrow focus potentially limits the exploration of novel MMCL methods because their key motivation and techniques are similar to methods in unimodal CL settings.

B. Architecture-Based Approach

Architecture-based methods employ an intuitive and direct strategy to learn tasks by enabling different model parameters to cope with different tasks. Regularization-based methods share all parameters to learn tasks, making them prone to intertask interference [5]: an issue where remembering old tasks greatly interferes with learning a new task, leading to decreased performance, when the forward knowledge transfer is negative [110]. In contrast, architecture-based methods reduce intertask interference by incorporating task-specific components. Depending on the factors driving architectural designs, we categorize methods into two types: *task-driven architecture* and *modality-driven architecture*. We provide an

TABLE I
SUMMARY OF MMCL METHODS FOR VISION AND LANGUAGE

	Method	MMCL Scenario					Task	MM Backbone	CL-			CA				Code
		CIL	DIL	XDIL	TIL	MDTIL			L	V	MI	C1	C2	C3	C4	
Regularization	CR	TIR [78]			✓		GEN	BLIP2 [102], InstructBLIP [103]	✓		✓					-
		SCD [32]				✓	CLS	ViLT [38]	✓	✓	✓					-
		CS-VQLA [79]	✓				CLS	VisualBERT [104]	✓	✓	✓					Link
		SnD [77]			✓	✓	CLS	CLIP [67]		✓					•	Link
	RR	ZSCL [55]	✓			✓	CLS	CLIP	✓	✓			•		•	Link
		Mod-X [49]	✓				RET	CLIP	✓	✓			•			-
		CTP [80]	✓				RET	-	✓	✓	✓		•			Link
		MSPT [43]	✓				CLS	-	✓	✓	✓	•	•			Link
		MuKI [81]	✓			✓	CLS	CLIP	✓	✓	✓		•		•	-
Architecture	TA	RATT [82]			✓		GEN	-	✓	✓	✓					Link
		MoE-Adapters4CL [52]	✓			✓	CLS	CLIP*	✓	✓				•	•	Link
		DIKI [85]	✓				CLS	CLIP*	✓	✓				•	•	Link
		CLAP [51]	✓				CLS	CLIP	✓		✓		•		•	Link
		VLKD [50]				✓	RET	-	✓	✓	✓		•			-
		EProj [78]			✓		GEN	BLIP2, InstructBLIP			✓			•	•	-
		MSCGL [83]	✓				CLS	-	✓	✓	✓		•			-
		SS [84]				✓	RET	-			✓		•			Link
	MA	SCML [66]				✓	CLS, RET	-	✓	✓	✓	•	•			-
Replay	NR	TAM-CL [86]				✓	CLS	-	✓	✓	✓					Link
		VQACL [33]			✓		GEN	-	✓	✓	✓					Link
		KDR [87]		✓			RET	-	✓	✓	✓		•			-
		GMM [88]			✓		CLS	BLIP2		✓				•	•	Link
		RAPF [89]	✓				CLS	CLIP		✓				•	•	Link
		RAIL [76]			✓		CLS	CLIP		✓				•	•	Link
	IR	IncCLIP [90]	✓	✓			CLS, RET	CLIP	✓	✓	✓		•			-
		SGP [73]			✓		GEN	-		✓	✓		•			Link
		ConDU [92]			✓	✓	CLS	CLIP	✓	✓					•	-
	MP	CPE-CLIP [53]	✓				CLS	CLIP*	✓	✓			•	•	•	Link
		TRIPLET [94]				✓	CLS	ALBEF [105], FLAVA [106]*	✓	✓	✓		•	•	•	-
		S-liPrompts [93]		✓			CLS	CLIP*	✓	✓				•	•	Link
		HPC [44]	✓				RET	CLIP, GLIP [107]*	✓	✓			•	•	•	-
		DPeCLIP [95]			✓		CLS	CLIP*	✓	✓	✓			•	•	Link
		MM-Prompt [46]			✓		GEN	LXMERT [108] + T5 [109]*	✓	✓	✓	•	•	•	•	Link
Prompt	UP	Fwd-Prompt [54]			✓		GEN	BLIP2, InstructBLIP*	✓	✓	✓			•	•	-
		CoLeCLIP [96]			✓	✓	CLS	CLIP*	✓	✓				•	•	Link

“MMCL Scenario”: defined in Section II-C; “MM Backbone”: the MM backbone of the MMCL methods, and ★ indicates that the model uses parameter-efficient fine-tuning (PEFT) strategies; “Task”: *CLS* means classification for input modality (or modalities), *RET* means image-text retrieval, and *GEN* means text generation; “CL-V/L/MI (Vision/Language/Modality Interaction)”: ✓ indicates that the model continually learns vision information, language information, and modality interaction, respectively; “CA (Challenges Addressed)”: • indicates challenges 1 to 4 described in Section I that the method has addressed. Note that because all methods aim to address challenge 0, we omit it in the table to enhance clarity and emphasize efforts towards new MMCL challenges; “Code”: the open-source implementation. “-” represents non-existence.

TABLE II
SUMMARY OF MMCL METHODS FOCUSING ON MODALITIES OTHER THAN VISION AND LANGUAGE

	Method	MMCL Scenario					Modality						Task	CA				Code
		CIL	DIL	XDIL	TIL	MDTIL	Vision	Language	Audio	Tactility	Acceleration	Gyroscope		C1	C2	C3	C4	
Architecture	TA	CMR-MFN [37]	✓				✓					✓	CLS		•			Link
	MA	ODU [36]				✓	✓		✓	✓			CLS	•	•			-
Replay	NR	SAMM [31]	✓	✓			✓		✓				CLS		•			Link
	IR	AID [47]	✓				✓				✓	✓	CLS	•	•			-
		FGVIRs [40]	✓				✓				✓	✓	CLS	•	•			-
		exFeCAM [91]	✓				✓			✓			CLS	•				-

“Modality”: ✓ indicates that the respective modality is included.

overview of representative architectures for these two subcategories in Fig. 6(b).

1) Task-Driven Architecture: Task-driven architecture methods design mechanisms in response to new tasks in the MMCL setting. By dedicating different parameters to individual tasks, these methods inherently reduce intertask interference and mitigate forgetting. Typical mechanisms include task-specific parameter allocation within fixed networks and dynamic expansion of new modules. This architectural flexibility allows some methods to effectively

address *Challenge 2* through specifically designed modules. Furthermore, by freezing the pretrained multimodal backbone and only training lightweight new modules, some approaches also address *Challenges 3* and *4*.

It is worth noting that, if the model has task-specific components and receives the task-ID during testing (the TIL scenario), the primary objective remains to avoid forgetting; however, the model should also effectively learn shared knowledge across tasks and balance performance with computational complexity [4].

a) *MMCL Methods*: When a new task is introduced, one straightforward strategy is to include task identity as input to control parameter allocation or directly add a new module into the network, i.e., *direct task-driven*. A more sophisticated approach is to design a mechanism that adaptively determines how to modify the network for learning new knowledge while maintaining computation efficiency, i.e., in an *adaptive task-driven* manner. Therefore, we group methods based on these two mechanisms in the subsequent paragraphs.

In *direct task-driven* MMCL methods, tasks and task-specific parameters have a *direct* correspondence. RATT [82] introduces a binary vocabulary mask to selectively inhibit neurons and allocate distinct activations across layers for different tasks, inspired by HAT [111]. Alternatively, instead of operating on fixed models, adding modules increases model capacity with each new task, thereby ensuring that performance is not ultimately constrained by the initial capacity [50]. Both MoE-Adapters4CL [52] and DIKI [85] freeze the pretrained CLIP model and introduce task-specific parameter-efficient modules (Challenges 3 and 4). MoE-Adapters4CL contains LoRA [112] modules as experts within the MoE framework [113], along with task-specific routers that determine the weighted aggregation of experts. DIKI learns task-specific key and value matrices with a residual mechanism. When the task-ID is unknown, both methods design distribution-based mechanisms to infer the task and select the most relevant modules.

CMR-MFN [37] fixes encoders of each modality and adds a modality fusion network for each task in training. SS [84] also uses task-specific modules, such as feed-forward and attention blocks, but proposes different selective specialization strategies to select these modules and analyzes their effectiveness. CLAP [51] introduces a visual-guided attention module to align learned text features and pretrained image features. Moreover, it proposes task-specific adapters to capture task-specific text feature distributions, trained with probabilistic fine-tuning.

Some methods *adaptively decide* when to expand, prune, or alter the network during the training process. These methods mitigate the increased training costs and redundancy caused by simply adding network parameters for each task. EProj [78] is proposed alongside TIR [78] (introduced in Section III-A1), which leverages task similarity scores to determine whether to add a new task-specific module. If all similarity scores are below a threshold, EProj expands the projection layer in the multimodal base model for the new task, learns task-specific keys, and freezes other modules to prevent forgetting. VLKD [50] constructs a hierarchical recurrent network that expands to learn new knowledge and adaptively deletes less relevant parameters. MSCGL [83] is a multimodal graph model with structure-evolving graph neural network (GNN) cells, extending the framework of GraphNAS [114]. In the search space of aggregation, activation, and correlation operators, MSCGL aims to find the best architecture to learn new tasks.

2) *Modality-Driven Architecture*: Modality-driven architecture methods introduce structural changes in response to shifts in modalities, a scenario unique to MMCL. Unlike other methods that usually assume a fixed set of modalities, this category considers a more realistic situation where the

modalities of tasks may vary during the CL process. By explicitly designing mechanisms for dynamic modality sets, these methods effectively handle Challenge 1 and also offer novel ways to manage Challenge 2.

a) *MMCL Methods*: To handle changes in input modalities, SCML [66], as a *modality-driven* method, proposes an architecture as a unified model with a meta-learner and dynamic modality encoders.

In this framework, they propose so-called plug networks as dedicated encoders for individual modalities, which map features to the same dimension. It learns each modality sequentially and utilizes a meta-learner to update the unified model to avoid forgetting. Therefore, it is extensible for accommodating the arrival of new modalities. The advantage of SCML is that the unified model maps different modalities into a common feature space and avoids explicit alignment between modalities. ODU [36] develops classifiers for each task and modality. The data of some modalities may be available in the first task but missing in later tasks. It even trains classifiers of missing modalities, leveraging other modalities as the auxiliary information source.

b) *Summary with Pros and Cons*: In Fig. 6(b), we illustrate the typical architectures of task-driven and modality-driven architecture-based methods. To add CL capabilities to multimodal models, designing various architectures is an intuitive strategy and yields various advantages. First, methods reduce intertask interference with task-specific parameters and enhance learning capability. Another important advantage of this category is its flexibility and versatility, with methods designed to address each of the core MMCL challenges in Section I. Especially when applied to foundation models, an effective strategy is to freeze the pretrained encoders while only training lightweight modules [Fig. 6(b)]. Methods adopting this strategy (e.g., MoE-Adapters4CL, DIKI, and EProj) effectively reduce computational costs and preserve zero-shot capabilities (Challenges 3 and 4).

However, extra architectural designs introduce significant drawbacks. First, the model size grows with the arrival of new tasks, leading to increased computation costs and ultimately negating the expected efficiency of CL. A typical example is CLAP, whose trainable parameters could potentially exceed those of the pretrained CLIP [51]. Second, in CIL scenarios, methods such as MoE-Adapters4CL that rely on task-specific modules often require task-ID discriminators; such designs are, therefore, fragile, as discriminators' ineffectiveness directly leads to poor performance on corresponding tasks [95].

C. Replay-Based Approach

Replay-based methods utilize an episodic memory buffer to replay historical instances, such as data samples, from previous tasks, helping to maintain early knowledge while learning new tasks. This approach of replaying instances avoids the rigid constraints of regularization-based methods and circumvents the complexity of dynamically modifying network architectures in architecture-based methods.

Depending on the strategies to preserve and utilize multimodal information during the replay process, replay-based

methods are divided into two subdirections: *natural multimodal replay* and *interactive multimodal replay*.

When learning the t th task, the episodic memory $\mathcal{M}_t$ will be combined with the incoming data $\mathcal{D}_t$. The loss function can be expressed as

$$\mathcal{L}_t = \frac{1}{|\mathcal{D}_t \cup \mathcal{M}_t|} \sum_{(\mathbf{x}_i, y_i) \in (\mathcal{D}_t \cup \mathcal{M}_t)} \ell(f(\mathbf{x}_i), y_i). \quad (15)$$

We depict the representative architectures of these two subcategories in Fig. 6(c).

1) *Natural Multimodal Replay*: Natural multimodal replay aims to preserve the natural and inherent relationships between modalities within datasets. A straightforward approach is to directly replay stored multimodal data pairs or tuples. For tasks with a looser modality link (e.g., connected only by a shared label space rather than being uniquely paired), some methods generate data for each modality separately. The natural multimodal relationship is, thus, preserved implicitly through the same class label. This generative approach, as an extension of unimodal replay, also avoids the storage requirements and privacy concerns of replaying raw samples. Methods in this category primarily target *Challenge 0* through replay of past data information.

a) *MMCL Methods*: With multimodal data, an intuitive implementation involves directly selecting and replaying samples from various modalities.

For instance, following the sampling strategies from [115] and [116], VQACL [33], GMM [88], and SAMM [31] select multimodal samples randomly. The experimental results from SAMM [31] demonstrate that, compared to unimodal replay, multimodal replay significantly enhances the plasticity and stability of the model, thereby achieving a superior stability–plasticity tradeoff.

Once multimodal data are stored in a buffer, some methods naturally integrate with KD, ensuring that the model maintains consistency in various aspects of the old data before and after model updates [5]. To ensure consistency at the *representation level*, TAM-CL [86] computes the KD loss between the outputs of the last self-attention block from the current student model and the earlier teacher model. This strategy helps to constrain distribution shifts. In terms of consistency in *cross-modal interactions*, KDR [87] utilizes KD to regulate the cross-modal similarity matrix, thereby enhancing the consolidation of cross-modal knowledge.

When one modality contains a small fixed set of data and is always available, such as templated sentences for classes in CLIP-based image classification, methods focus on replaying the more complex modality. RAIL [76] employs recursive ridge regression with a memory matrix, theoretically achieving absolute memorization on visual knowledge. RAPF [89] generates old class image features and encourages them to be closer to the text embeddings of old classes and away from those of new classes, thereby mitigating forgetting in CIL.

2) *Interactive Multimodal Replay*: In contrast to natural replay, which preserves inherent multimodal relationships, interactive multimodal replay methods actively learn and utilize modality interaction information when generating data or

features for pseudo-replay. These methods typically capture this interaction knowledge through designed modules and prototypes. This category is particularly effective for addressing *Challenge 2*. It may also help alleviate *Challenge 1*, especially when complex interactions are involved with more than two modalities.

a) *MMCL Methods*: With datasets that include various modalities, generating highly correlated data tuples, such as image–question–answer triplets that are both detailed and accurately labeled, usually poses significant challenges. To address these difficulties, some studies have focused on learning modality interactions, thus effectively storing multimodal knowledge and generating either substitute or partial data. For instance, SGP [73] maintains scene graphs, which are graphical representations of images, and incorporates a language model for pseudo-replay. IncCLIP [90] emphasizes pseudo-text replay through the generation of negative texts conditioned on images, which helps better preserve learned knowledge. Within the robot operating system framework, exFeCAM [91] maps tactile and vision data to the same feature space and generates classwise prototypes for continual object recognition. ConDU [92] stores task-specific triggers with whole model updates and class prototypes to recover the model and perform prediction at inference. In addition, efforts like FGVIRs [40] and AID [47] specifically tackle issues of modality imbalance. They employ pseudo-representation and pseudo-prototype replay strategies to enhance classifier discriminability. They address the inherent challenges in multimodal learning environments where maintaining balance across different types of data is crucial.

b) *Summary with Pros and Cons*: As shown in Fig. 6(c), within the MMCL setting, replay-based methods offer great flexibility in deciding replay data, as they may opt to replay one or multiple modalities based on the specific design of the model. Furthermore, the required architectural changes to learn multimodal knowledge and facilitate replay are typically minimal compared to those in architecture-based methods.

However, there are limitations to these methods. First, for many methods in natural multimodal replay, directly utilizing past data induces privacy concerns when the data are sensitive. Moreover, the reliance on a memory buffer introduces a difficult performance–memory tradeoff: a fixed-size buffer often leads to performance degradation over long task sequences, while allowing the multimodal samples in the buffer to grow with tasks results in memory overhead [44], [93]. Lastly, some methods generate replay data based on specific data representations (e.g., scene graphs for SGP), which may not be readily available in practice, thus limiting their applicability [94].

D. Prompt-Based Approach

With the rapid development of large models and their application in the CL setting, prompt-based methods have recently emerged to better utilize the rich knowledge acquired during pretraining. These methods offer the advantage of requiring minimal model adjustments and reducing the need for extensive fine-tuning, unlike previous methods that often require significant fine-tuning or architectural modifications.

The paradigm of prompt-based methods involves modifying the input by applying a few prompt parameters in a continuous space, allowing the model to retain its original knowledge while learning additional task-specific information. Consequently, they are inherently capable of addressing *Challenges 3*, and *4* in the MMCL setting. Depending on the prompt design strategies, prompt-based methods are categorized into two types: *multimodal prompt* and *universal prompt*. We present representative architectures of these subcategories in Fig. 6(d).

1) *Multimodal Prompt*: Multimodal prompt methods are designed to learn prompts for each modality, extending from unimodal methods [65], [117], but they usually go beyond simply learning prompts in parallel. Importantly, methods explore the connections between these prompts during continual training, addressing *Challenge 2*.

a) *MMCL Methods*: Existing multimodal prompt-based works operate at various granularities, such as *task-specific* prompts (S-liPrompts [93]) and *layer-specific* prompts (CPE-CLIP [53] and TRIPLET [94]). These approaches place greater emphasis on designing prompts that cater to different modalities. For instance, S-liPrompts introduces a joint language-image prompting scheme that enables the image-end transformer to seamlessly adapt to new domains, while enhancing the language-end transformer's ability to capture more semantic information. DPCLIP [95] also learns separate prompts for the image and language branches, generating them via dedicated attention modules. Moreover, CPE-CLIP, TRIPLET, MM-Prompt [46], and HPC [44] focus more on explicitly considering multimodal interactions during continual prompt learning (*Challenge 2*). CPE-CLIP connects language and vision prompts by defining vision prompts as a function of language prompts; TRIPLET proposes decoupled prompts and prompt interaction strategies to model the complex modality interactions. HPC designs calibration mechanisms to enhance the consistency alignment of prompts across modalities.

2) *Universal Prompt*: Instead of learning prompts for different modalities separately, universal prompt methods aim to learn prompts that encapsulate shared knowledge, either across different modalities or at the task level. This strategy inherently avoids inconsistency among modality-specific prompts. Moreover, these methods offer a more computationally efficient option while still achieving competitive performance.

a) *MMCL Methods*: Fwd-Prompt [54] maintains a single prompt pool shared across all modalities and tasks, and dynamically chooses the most relevant ones based on inputs. CoLeCLIP [96] utilizes task prompts to capture task-specific features and introduces an attention mask to avoid interference between prompts and the model's pretrained features.

b) *Summary with Pros and Cons*: We summarize the key architectures of multimodal and universal prompt-based methods in Fig. 6(d). In the MMCL setting, prompt-based methods may choose to modify the input and learn prompts for the encoders of modalities and/or the modality interaction component. Depending on the model design, these methods may also facilitate interactions between prompts across different modalities. As a newly emerged paradigm built upon foundation models, the prompt-based approach offers several

distinct advantages. The primary strength is that it successfully preserves the pretrained knowledge and learns downstream tasks parameter-efficiently (*Challenges 3* and *4*). Moreover, prompts are flexible to be applied at different granularities, enabling the acquisition of multiperspective and comprehensive knowledge.

However, this category also presents unique limitations. First, unlike hard prompts with human-readable words, soft learnable prompts lack explainability, making it difficult to understand how they evoke pretrained knowledge. Second, their effectiveness relies heavily on the underlying pretrained models, unlike methods in other categories, which also work with models trained from scratch. Third, prompts with restricted length limit their learning and expressive capabilities [92]. Lastly, for methods like S-liPrompts and CoLeCLIP, the prompt parameters scale linearly with the number of tasks, raising concerns in computational and memory costs for long task sequences (*Challenge 3* reintroduced) [96].

IV. DATASETS AND BENCHMARKS

In this section, we provide an overview of current datasets and benchmarks in MMCL.

There are three primary construction approaches: 1) collecting new datasets that are dedicated to MMCL; 2) linking multiple well-known datasets that are initially designed for non-CL tasks; and 3) partitioning a single dataset into multiple subsets to simulate tasks in the MMCL setting [42].

Table III summarizes MMCL benchmarks covering various CL scenarios, modalities, and task types. We introduce them as follows if the codes are publicly accessible.

A. Benchmarking on an Original Dataset

In this section, we summarize three benchmarks with newly collected datasets dedicated to the MMCL setting. Compared to the other two approaches, which manually alter existing datasets, these original datasets help evaluate and scale for more practical MMCL settings. Zhu et al. [80] utilize E-commerce data to construct the first vision-language continual pretraining dataset P9D and establish the VLCP benchmark for cross-modal retrieval and multimodal retrieval. P9D contains more than one million image-text pairs of real products and is partitioned into nine tasks by industrial categories. Ahrens et al. [84] introduce the LILAC benchmark with two datasets: LILAC-2D generated based on minigrid [118] and LILAC-3D based on Ravens [119] and CLIPort [120], for language-instructed tasks grounded in simulated visual environments. Baseline CL methods (ER [115] and EWC [16]) under LILAC-2D yield suboptimal performance, with an average accuracy more than 20% lower than the newly proposed MMCL method SS [84]. Xu et al. [42] collect video and sensor data from ten participants wearing smart glasses. They construct the dataset UESTC-MMEA-CL, the first multimodal dataset for continual egocentric activity recognition, with modalities of vision, acceleration, and gyroscope. They also establish a benchmark, CEAR, with three baseline CL methods, namely, EWC [16], LwF [15], and iCaRL [17]. Results demonstrate that replay-based iCaRL is more

TABLE III
SUMMARY OF MMCL BENCHMARKS

Name	MMCL Scenario					Modality					Task	Code
	CIL	DIL	GDIL	TIL	MDTIL	Vision	Language	Audio	Acceleration	Gyroscope		
CLiMB[39]				✓	✓	✓	✓				CLS	Link
CLOVE [73]			✓			✓	✓				GEN	Link
IMNER, IMRE [43]	✓					✓	✓				CLS	Link
MTIL [55]				✓		✓	✓				CLS	Link
VLCP [80]	✓					✓	✓				RET	Link
LILAC [84]				✓		✓	✓				RET	Link
MMCL [31]	✓	✓				✓		✓			CLS	Link
CEAR [42]	✓					✓			✓	✓	CLS	Link

effective in alleviating forgetting than replay-free methods, EWC and LwF. Nonetheless, exploring replay-free strategies remains promising and important, as replay-based methods are not always applicable due to considerations such as privacy concerns [42]. Xu et al. [42] use TBW-like [121] midfusion to fuse multimodal features, achieving better results than using single modality data in the non-CL setting. However, in the MMCL setting, the performance with multimodal data (vision and acceleration) is inferior to that with unimodal data (vision), even with CL methods incorporated. These results highlight the necessity for further research in MMCL methods to improve the fusion of modality information while preventing forgetting.

B. Benchmarking on Several Datasets

We outline three benchmarks that utilize various existing datasets as tasks in the MMCL framework. These independently collected datasets exhibit drastically different distributions and are challenging for model learning. CLiMB [39] benchmarks with four vision–language tasks (VQAv2 [72], NLVR2 [122], SNLI-VE [74], and VCR [123]), five language-only tasks (IMDb [75], SST-2 [124], HellaSwag [125], CommonsenseQA [126], and PIQA [127]), and four vision-only tasks (ImageNet-1000 [128], iNaturalist2019 [129], Places365 [130], and MS-COCO object detection [131]). CLiMB treats each task as a classification task and consists of two phases within the CL process. In upstream CL, the model is trained on vision–language tasks with various candidate CL algorithms. In downstream low-shot transfer, after training on the i th upstream task and saving checkpoints, for each task of the training data of the remaining upstream tasks and unimodal tasks, the model is fine-tuned on the checkpoints with a fraction of the task data. The CLiMB benchmark results demonstrate that common CL algorithms (ER [115] and EWC [16]) are able to alleviate forgetting. However, they may hurt downstream task learning, compared to direct fine-tuning. These results underscore the need for further research on MMCL methods. CLOVE [73] splits data from GQA [35] into six subsets representing different scenes, such as *workplaces* for the CLOVE-scene CL setting, following the taxonomy in SUN397 [132]. Additionally, CLOVE collects six functions, such as *object recognition*, for the CLOVE-function CL setting, using data from GQA [35], CRIC [133], and TextVQA [134]. CLOVE evaluates the performance of different methods on the CL of different VQA tasks. Lastly,

MTIL [55] is a challenging benchmark consisting of 11 image classification tasks from different domains, including Aircraft [135], Caltech101 [136], CIFAR100 [70], DTD [137], EuroSAT [138], Flowers [71], Food [68], MNIST [139], OxfordPet [69], StanfordCars [140], and SUN397 [132].

C. Benchmarking on a Partitioned Dataset

In MMCL benchmarking, the most straightforward approach is to utilize a well-known dataset initially designed for non-CL settings and partition it into multiple subsets to simulate a sequence of tasks. Such simplicity facilitates rapid initial progress in MMCL research, with three benchmarks developed in this manner.

The IMNER benchmark [43] utilizes the Twitter-2017 MNER dataset (constructed by Lu et al. [141] and preprocessed by Yu et al. [142]) and splits it by categories to simulate the CIL scenario. The IMRE benchmark [43] partitions the MEGA MRE dataset [143] into ten subsets for the CIL scenario. MMCL [31] is a benchmark that contains audio and visual modalities for classification. It partitions the VGGSound dataset [144] to simulate CIL and DIL scenarios.

V. DISCUSSION

In this section, we present an in-depth discussion of research trends, foundation models in MMCL, and the evolved stability–plasticity tradeoff.

A. Observation of Research Trends

As MMCL methods continue to emerge, we observe and summarize two pivotal research trends. The first trend is a significant shift in the challenges that MMCL methods focus on, which has progressively distinguished MMCL from traditional CL and fostered the development of this field. The second trend is a transition from training models from scratch to adapting pretrained foundation models, providing stronger CL capabilities and better practical utility in real-world scenarios.

1) *Challenges*: Inherited from traditional unimodal CL, whose major challenge is catastrophic forgetting, early MMCL methods naturally only focus on Challenge 0 (see [82]). Subsequently, MMCL works start to identify and address new challenges in this field (Section I). This evolution has advanced MMCL as a unique research area, differentiating it from traditional CL. Consequently, many recent works

address these new challenges, yielding improved performance and alleviating the more severe forgetting that occurs in the multimodal setting [31], [42].

2) *Backbones*: Mirroring a broader trend in the ML community, backbones in MMCL have evolved from small, randomly initialized models to large foundation models. Many early approaches design custom architectures and train them from scratch (see [36], [50], [82]). The rise of foundation models has profoundly changed the landscape of the field, with a majority of recent MMCL methods leveraging them as powerful backbones (Table I). This adoption equips models with a head start in pretrained knowledge, greater learning capacity, and better performance. We discuss further details in the next section.

B. Foundation Models in MMCL

In this section, we explore the pivotal role of foundation models and discuss MMCL methods based on them.

1) *Foundation Models*: Pretrained on large-scale multimodal datasets, foundation models such as CLIP [67], BLIP2 [102], and GLIP [107] have acquired extensive general knowledge. Performing CL on top of these MM backbones, thus, provides a significant head start. Simultaneously, these foundation models cannot remain static but require CL to adapt to the dynamically evolving world. Full retraining with both pretrained and new data is computationally prohibitive, underscoring the necessity and importance of MMCL. This establishes a mutually beneficial relationship between foundation models and MMCL methods.

Importantly, foundation models offer remarkable flexibility as backbones, as they are inherently multimodal. Besides those pretrained on multimodal data, even unimodal-trained models, such as LLMs, can be transformed into powerful multimodal learners (MLLMs). By extracting and feeding features from other modalities, LLMs can adapt to multimodal tasks. Consequently, some MMCL methods do not directly use off-the-shelf MM backbones but instead assemble custom ones (e.g., MM-Prompt [46] combines LXMERT [108] and T5 [109]). The selection and combination of foundation models in MMCL are diverse, fostering a flourishing of various MMCL techniques built upon them, which we detail next.

2) *MMCL Methods*: The rise of foundation models has enriched the landscape of MMCL, with a primary focus on vision–language modalities (Table I). At the *task level*, the choice of backbone often links to the task type. Many CLIP-based MMCL methods focus on image classification, which involves matching images with templated class sentences from multiple domains (see [52], [55], [76]). Meanwhile, MLLM-based approaches are frequently applied to generative tasks like continual VQA (see [33], [46], [73]). At the *technique level*, methods focus on preserving and utilizing knowledge of foundation models when learning new knowledge. They protect existing knowledge at the parameter level (regularization-based) and the data level (replay-based). Meanwhile, some techniques aim to elicit knowledge from foundation models through learned prompts (prompt-based) and facilitate new knowledge acquisition through architectural modifications (architecture-based). Prompt-based category, in

particular, is a newly emerged paradigm whose design is uniquely suited to powerful foundations. Concurrently, other established categories also further advance with new methods designed for these foundations.

3) *Evaluations*: While the diversity of foundation models facilitates the development of MMCL methods, as discussed in Sections V-B1 and V-B2, it highlights an evaluation issue. For a fair comparison, methods should be adapted to and evaluated on the same backbone. We note that for some datasets, the performance ranking of methods can invert with backbone changes (e.g., EProj [78] versus Fwd-Prompt [54] on BLIP2 [102] and InstructBLIP [103]). This phenomenon shows the backbone dependence of these foundation model-based methods and highlights the need to enhance robustness.

C. Stability–Plasticity Tradeoff

In this section, we present an in-depth discussion of the stability–plasticity tradeoff dilemma, with an emphasis on its evolution under MMCL methods with foundation models.

1) *Stability*: Stability refers to a model’s ability to retain previously learned information [12]. For MMCL methods with small models trained from scratch, stability primarily concerns knowledge acquired from the task sequence, as randomly initialized models begin with no prior knowledge. However, the incorporation of foundation models expands the scope of stability to include a crucial second dimension: preserving their pretrained general knowledge (Challenge 4). Forgetting pretrained knowledge in training leads to loss of *zero-shot capabilities* and degrades performance on future tasks, i.e., negative FWT [54], [55]. On the other hand, while simply using the original pretrained checkpoint ensures perfect stability, it is inadequate because direct fine-tuning and MMCL methods allow new knowledge acquisition and achieve better performance on the current task.

Notably, foundation models possess a high degree of inherent stability and demonstrate mitigated forgetting. Even after direct fine-tuning on new tasks, despite performance degradation, they often maintain the performance of old tasks at a reasonable level [55], rather than regressing to near-random guessing like small models [16]. Their vast general knowledge and large number of parameters prevent a complete loss of knowledge. Therefore, building MMCL methods on foundation models provides a more robust starting point, which aligns with the research trend discussed in Section V-A.

2) *Plasticity*: Plasticity is defined as the capacity to acquire new knowledge [12]. While small models are computationally inexpensive, they face large limitations due to their model capacity, especially with long task sequences. Large foundational models are better at adapting to incoming tasks with pretrained knowledge and a large number of parameters. However, full-parameter fine-tuning comes at a high training cost, posing practical concerns (Challenge 3). Consequently, some MMCL methods focus on controlled adaptation with a manageable number of trainable parameters. By freezing most parameters and training only a small subset, these methods achieve plasticity and address Challenge 3.

3) *Tradeoff*: Plasticity enables the acquisition of new information, but it also causes disruption, leading to forgetting.

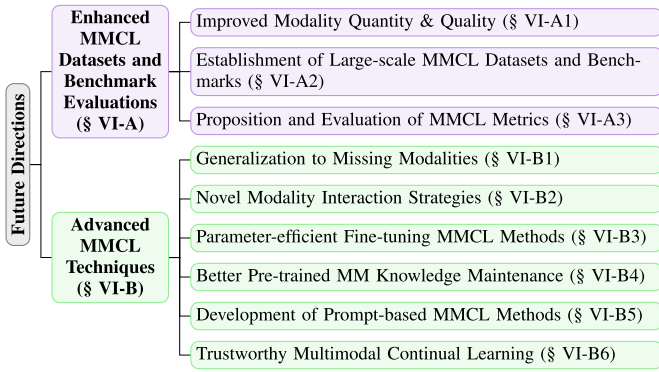


Fig. 7. Taxonomy of future directions of MMCL.

Conversely, excessive stability, while preventing forgetting of previous tasks, is undesirable if it impedes adaptation [12]. A tradeoff between the two is, therefore, necessary. For small models with unimodal data, methods such as EWC [16] and progressive network [145] address this issue by regularizing on parameters or initializing new networks for each task. However, multimodal data inevitably increase the parameters required for learning. For large foundation models, these traditional strategies become challenging to scale due to computation and storage burdens. Therefore, more efficient MMCL solutions are needed to achieve this balance.

With foundation models, the situation is further complicated by the need to manage an additional tradeoff between plasticity and the stability of pretrained knowledge, which gives powerful zero-shot generalization capabilities [81], [85]. Some methods achieve this balance by freezing most parameters or keeping a copy of the original model, while using a small set of trainable parameters to learn new knowledge. The key idea in modern methods for managing this new tradeoff is to decouple general and task-specific knowledge. Architecture and prompt-based methods achieve this explicitly by design, storing new knowledge in dedicated modules or prompts. In contrast, regularization and replay-based methods, without additional strategies, typically require full-parameter fine-tuning and entangle these two knowledge types within a shared parameter space, leading to intertask interference.

Beyond the overall multimodal tradeoff, MMCL should also consider the tradeoff for each modality. Due to modality imbalance (Challenge 1), modalities do not learn or forget in a synchronized manner [40]. For instance, language features naturally dominate vision features in continual VQA [46]. This crucial aspect is currently addressed by only a few MMCL methods (see [43], [46]).

In summary, while various methods have been proposed to manage the evolved stability–plasticity tradeoff, MMCL is still in its early stages. This dilemma remains a critical issue that requires further research and development.

VI. FUTURE DIRECTIONS

With the rapid advancement of multimodal models, MMCL has become an active and promising research topic. As illustrated in Fig. 7, we outline future research directions along two primary areas: 1) the development of more comprehensive

datasets and benchmark evaluations and 2) the design of novel and effective MMCL techniques.

A. Enhanced MMCL Datasets and Benchmark Evaluations

1) *Improved Modality Quantity and Quality*: Our summarization in Table II reveals that only a few MMCL methods focus on modalities other than vision and language. Therefore, there is a huge space for further research on incorporating more modalities. Similarly, developing benchmarks for more modalities is important for this field. Moreover, modalities are not limited to those listed in Table II and may include biosensors [146], genetics [147], and others [148], thereby enhancing support for emerging challenges in fields such as AI for science research. Various multimodal methods and applications [149], [150], [151], [152] can be extended into the CL framework to enhance their learning capabilities. With the introduction of more modalities, it will be increasingly imperative to address *data-level modality imbalance*, i.e., Challenge 1, which, as shown in Tables I and II, has been addressed by only a few MMCL methods. Furthermore, due to the discrepancy among distributions and quality of different modalities, the modality with better performance may dominate optimization, leaving other modalities under-optimized [45]. Hence, addressing *parameter-level modality imbalance* is also crucial. Developing specific strategies to balance modalities helps mitigate the forgetting issue [40], making it a promising research direction.

2) *Establishment of Large-Scale MMCL Datasets and Benchmarks*: Many current MMCL datasets are constructed by simply linking or adapting existing academic datasets, as mentioned in Section IV. While these datasets are relatively straightforward to set up and facilitate rapid initial progress for MMCL research, their limitations in scalability and representativeness become apparent as the field develops. These datasets often lack the scale and real-world complexity necessary for studying practical MMCL settings. Specifically, real-world data are long-tail distributed while manually constructed datasets typically have balanced classes or the same sample size for different tasks [80]. Moreover, improper data partitioning strategies due to a lack of information may not effectively simulate the continual environment [80]. Consequently, the need for new large-scale datasets for MMCL is increasingly urgent. Although such dedicated datasets are currently scarce, they already provide important insights through benchmark results, as summarized in Section IV-A. Therefore, we advocate for the establishment of more large-scale MMCL datasets and benchmarks as an important future direction, which will significantly advance MMCL research.

3) *Proposition and Evaluation of MMCL Metrics*: The new MMCL challenges require dedicated, fine-grained metrics for a deeper analysis of model capabilities and limitations. For initial modality-oriented metrics that have been proposed (Section II-D), we advocate for their wider adoption and systematic evaluation across more methods. Moreover, we call for the development of additional metrics to analyze the degree of challenge resolution from multiple perspectives. The development and evaluation of MMCL metrics will further advance this field from its early stage to a more mature phase.

B. Advanced MMCL Techniques

1) *Generalization to Missing Modalities*: A key real-world challenge is handling missing modalities (Challenge 1), which introduces practical difficulties to models' robustness. It may occur at the sample level for individual data points, or more severely, at the task level, where an entire task lacks a modality. Models require systematic designs to address this issue, without assuming the completeness of all modalities. Developing effective strategies to capture the intrinsic connection between tasks and modalities helps improve model performance [36]. A promising starting point for future research is to extend non-CL methods with compositional generalization capabilities, such as generating missing modality features [153], [154], to the CL scenarios.

2) *Novel Modality Interaction Strategies*: As we have just mentioned, there are only a few MMCL methods that incorporate more than two modalities. Modality interaction, especially modality alignment, may be more complicated with three or more modalities, i.e., Challenge 2. Furthermore, many existing MMCL methods simply fuse modalities within neural architectures without a deeper understanding or analysis of their mutual influence on learning. Thus, it will be interesting and promising to measure such intermodality influence [48], [155] for more fine-grained multimodal interaction.

3) *PEFT MMCL Methods*: Parameter-efficient fine-tuning (PEFT) methods offer an effective solution to optimize training costs, i.e., addressing Challenge 3, by reducing the number of trainable parameters while achieving comparable or better performance than full-parameter fine-tuning to the large models [112], [156]. While prompt-based methods are parameter-efficient, in Table I, we observe that in other categories, only a few methods utilize PEFT techniques. CLAP [51] also mentions this as its future work. Therefore, given numerous PEFT methods emerging in recent years [157], [158], employing them to reduce training costs for MMCL methods is a worthy direction. Furthermore, beyond the straightforward application of existing PEFT methods, a promising direction is to propose new PEFT methods specifically for the MMCL setting, and to seamlessly integrate them with other MMCL techniques.

4) *Better Pretrained MM Knowledge Maintenance*: As many MMCL methods are armed with powerful MM backbones, it is naturally desirable to memorize their pretrained knowledge during training. Forgetting pretrained knowledge may significantly hurt future task performance [54], [55]. We observe that few methods in Table I, aside from prompt-based ones, explicitly prioritize maintaining pretrained knowledge, i.e., addressing Challenge 4, as one of their key goals. Moreover, this is particularly challenging for replay-based methods that usually rely on quick adaptation to old data samples for knowledge retention. However, for certain pretrained models like CLIP, the pretrained data are private [55], which makes the target difficult yet promising for future research.

5) *Development of Prompt-Based MMCL Methods*: As discussed in Section III-D, prompt-based MMCL methods effectively address Challenges 3, and 4. However, as shown in Table I, we note that prompt-based MMCL methods are currently the least represented category. Recently, prompt

learning techniques are gaining traction in the non-CL setting for multimodal models [159], [160]. Moreover, there are popular prompt tuning methods that combine learning with high-quality templates [161]. Extending these methods to the MMCL setting facilitates the efficient and effective utilization of pretrained models. Given that the prompt-based category is still in its infancy, there is significant potential for further research and development.

6) *Trustworthy MMCL*: As public concern for model safety and privacy increases, and governments impose more related regulations, the demand for trustworthy models is escalating. To ensure safety, models should prevent and detect *hallucination* and *misinformation* to mitigate risks when deployed in the real world [162], [163]. Although there have been related studies in LLMs [164], such issues in MMCL have not been explored yet. A study on multimodal LLMs demonstrates that fine-tuning these models on a single dataset already causes hallucination [165]. While the model learns knowledge during the CL process, it should also continue to control and mitigate hallucination and misinformation. Thus, an important and interesting future direction is to extend such studies in MMCL. Moreover, models require *safety alignment* with human expectations and social standards to avoid misleading and harmful content [166]. However, the concept of safety changes and evolves over time. Thus, incorporating MMCL techniques into multimodal models is essential to address this changing concept. Regarding privacy, techniques such as federated learning (FL) enable the server model to learn knowledge of all clients' data without sharing their raw data. FL techniques also help enhance model robustness, keeping it stable even under extreme conditions such as malicious attacks intended to compromise the model [167], [168]. With numerous federated CL (FCL) methods, it would be a promising direction to extend FCL methods to the MMCL setting, so that models maintain knowledge of previous tasks and fuse new knowledge from multimodal data of clients with privacy protected [63].

VII. CONCLUSION

In this work, we present an up-to-date MMCL survey. We provide a structured taxonomy of MMCL methods, essential background knowledge, a summary of datasets and benchmarks, and discuss two novel MMCL scenarios for further study. We categorize existing MMCL works into four categories, i.e., regularization-based, architecture-based, replay-based, and prompt-based methods, with detailed subcategories described. We also provide representative architecture illustrations for all categories. Our detailed review highlights the key features and innovations of these MMCL methods. Additionally, we outline promising future research directions in this rapidly evolving field, offering discussions on potential areas for further investigation and exploration. We anticipate that the development of MMCL will further enhance models to exhibit more human-like capabilities. This enhancement includes the ability to process multiple modalities at the input level and acquire diverse skills at the task level, thereby bringing us closer to realizing general-purpose intelligence in this multimodal and dynamic world.

ACKNOWLEDGMENT

The icons are from PowerPoint and FlatIcon. The robots in Fig. 1 are generated by AI Image Creator in Bing. The authors sincerely thank Prof. Tat-Seng Chua at the National University of Singapore, for his invaluable and insightful suggestions regarding this article. They thank the anonymous reviewers for their constructive comments.

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